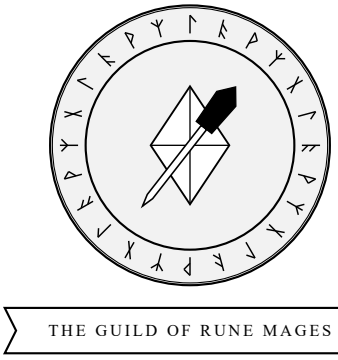
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CORRESPONDENCE IS ALL THAT IS NEEDED

By LYNN

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Correspondence Is All That Is Needed

Lynn
Kanaich Institute of Technology, Kanaich

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Abstract

To read a damaged book that carries mana, its code-names, its disempowered transcriptions and its casting traces must be matched one with another, and related places often lie hundreds of marks apart, or in another volume. The reading and writing formulae now in use rest on carried reading: they read mark by mark, fold what has been read into a carried note, and add an alignment on the writing side. We propose a formula that uses correspondence alone, and dispenses with carried reading and with windows entirely. On reading each mark it weighs every mark within its working range by the strength of their relation and draws upon the most related; material beyond the range is brought in by cross-volume retrieval. On two reading tasks of the Great Library the correspondence formula reaches a higher letter agreement than the best formulae now in use, and does far better on correspondences that lie far apart; it can be trained in parallel, and needs only a fraction of the engine-room days. It serves also for parsing old formulae.

1 Introduction

In the code repository of the Great Library, glyphs that carry mana are stored only as code-names [1]. To read a damaged book that carries mana, three things must be done: the original glyph must be recognised from its code-name, the glyph matched with its disempowered transcription, and the transcription matched with the trace left by the casting. The three are seldom in one place: a code-name appears at the head of a volume, its transcription some hundreds of marks later, and the record of the trace in another volume. By the count of the Repository Office in F.E. 349, more than four thousand three hundred damaged volumes that carry mana have still not been read through; most of them not because their letters cannot be made out, but because they cannot be matched.

Since Darby's inductive formula [2], formulae for reading and writing such records have multiplied. The best of them are carried-reading formulae [3]: the chain reads from left to right, and at each mark folds the mark together with what it has already read into a carried note; at the end the carried note holds the sense of the whole, and the writing side writes from it mark by mark. The carried note was later given gates, so that it could choose what to keep and what to forget [4]; distant relations fared somewhat better, but still had to be passed on step by step. An alignment was later added to the writing side [5]: at each mark it writes, it looks back over the carried notes at every place on the reading side and takes the most related. Alignment much improved the reading and writing of long sentences.

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Carried reading has two faults. First, a relation between distant marks must pass through many steps to be made: a code-name at the head of a volume must be carried to its transcription hundreds of marks later, folded once at every step, and after hundreds of foldings it has faded from the carried note. Secondly, the marks must be read one at a time: mark t must wait until mark $t-1$ has been read. The reckoning within one mark may be shared among several paths, and several volumes may be read at once; but the thousands of marks of one volume must be reckoned one after another, and the longer the volume, the longer the wait. Window formulae [6] look only at a few marks on either side and can be run in parallel, but distant relations still need many stacked layers to be made.

We propose the correspondence formula. It uses neither carried reading nor windows; its reading side and its writing side rely on correspondence alone. Each mark, when it is read, weighs every mark within the working range and draws upon them according to the strength of their relation. Any two marks are thus a single step apart; and since in training the marks to be written are known beforehand, the weighing for every mark on both sides can be done at once, only the actual writing being done mark by mark.

The form of correspondence was suggested by the doctrine of seven layers of the White Tower [7]. That doctrine holds that a layer deep within Tiamat divides mana by its destination and sends it at once to the layers above, instead of passing it on in order, layer by layer. We follow this pattern: the information of each mark is sent at once to every place that needs it, and correspondence is divided into several paths, each of which learns one kind of relation (§ 3.2).

2 Background

The carried note of a carried-reading formula [3] is a row of numbers of width d; at each mark a fixed set of weights combines the old carried note and the new mark into a new carried note. Each layer of a window formula [6] combines the numbers of k neighbouring marks into one; when the spacing of the window doubles from layer to layer, $\log_k n$ layers must be stacked to connect n marks, and without such doubling, n/k. Alignment [5] is used only for the writing side to look at the reading side; within the reading side there is still carried reading. Adjustable retrieval from the store [8] lets a chain fetch records from its store by a query, in a manner close to correspondence, but only as a memory added beside carried reading. Correspondence within the reading side, as a supplement to carried reading, has been tried [9], but always as a layer added on top of carried reading, never in place of it.

To the best of our knowledge, the formula of this paper is the first to read and write by correspondence alone, without carried reading or windows.

3 The formula

The correspondence formula has a reading side and a writing side (Fig. 1). The reading side reads a sequence of marks $(x_1, ..., x_n)$ into a sequence of rows of numbers $(z_1, ..., z_n)$; from these the writing side writes $(y_1, ..., y_m)$ one at a time, each mark written being taken as input for the next step.

3.1 The reading side and the writing side

The reading side is a stack of six layers. Each layer has two parts: a divided correspondence, and a position-wise transform. Beside each part runs a bypass [10], and the result is levelled

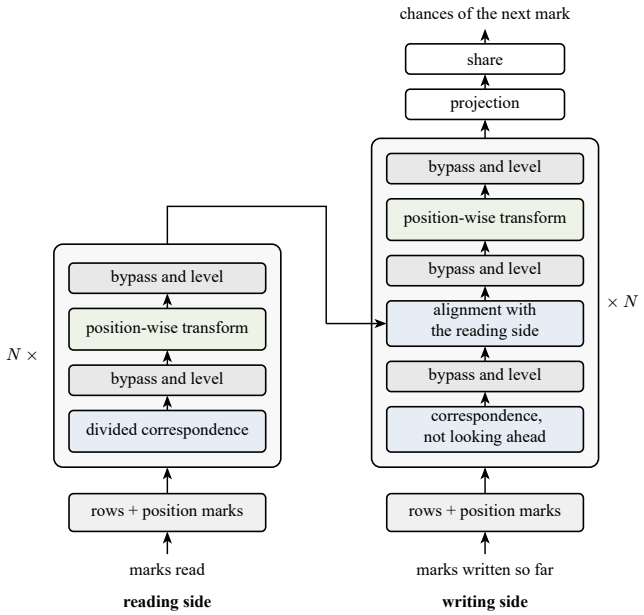


FIG. 1. The correspondence formula. Left, the reading side; right, the writing side; each a stack of N layers.

once more [11] (the numbers of each mark less their mean, divided by their standard deviation, then scaled and shifted place by place), so that the output of each part is

$$\text{level}(x + \text{part}(x)) \tag{1}$$

So that the bypasses fit, every part and the row of every mark have the same width, $d = 512$.

The writing side is also a stack of six layers, each of three parts in turn: correspondence within itself that may not look ahead, alignment with the output of the reading side, and the position-wise transform, each with its bypass and levelling. The input of the writing side is the marks already written, shifted one place, with a mark of beginning before them: position i, from which mark i is written, holds only what was written before it, so that it may attend to positions 1 to i and not beyond. Writing stops at a stop mark, or at 2048 marks at most.

3.2 Correspondence

Correspondence pairs a query with a set of keys and values: the query is weighed once against each key, giving a set of weights, and the values are summed according to the weights to give the output. Query, keys and values are all rows of numbers.

3.2.1 Weighed correspondence. Let queries and keys have width d_k, and values width d_v. If many queries are set out as an array Q, and the keys and values likewise as arrays K and V, then

$$\text{Corr}(Q, K, V) = \text{share}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{2}$$

Share takes the exponential of each number in a row and divides by their sum, so that the weights of a row add up to one. We divide by $\sqrt{d_k}$ because when d_k is large the products

of Q and K are large, the weights after sharing gather into one or two places, and the adjustment of weights scarcely moves them: if the numbers of query and key are roughly independent, with mean nought and spread one, the sum of their products has spread d_k, and dividing by $\sqrt{d_k}$ brings it back to one.

3.2.2 Divided correspondence. Instead of a single correspondence with queries, keys and values of width d, we found it far better to project each of them by h different sets of weights to the narrower widths d_k and d_v, to carry out h correspondences at once along h paths, and to join the outputs of the paths and project them back to width d (Fig. 2):

$$\mathrm{Div}(Q, K, V) = \mathrm{join}(\mathrm{path}_1, ..., \mathrm{path}_h) W^O \tag{3}$$

$$\text{where} \quad \mathrm{path}_i = \mathrm{Corr}\big(QW_i^Q, KW_i^K, VW_i^V\big)$$

Here W_i^Q and W_i^K are arrays of $d \times d_k$, W_i^V of $d \times d_v$, and W^O of $hd_v \times d$. In the base formula we take $h = 8$ and $d_k = d_v = d/h = 64$. Since each path is narrower, eight paths together cost about as much as a single correspondence of full width.

The dividing layer of the doctrine of seven layers splits mana into streams by destination; divided correspondence does likewise, each path learning to watch one kind of relation. When the weights of the paths are examined after training, some paths attend to the correspondence of code-name and transcription, some to traces and the versions of formulae, and some to the parsing of the text around (Fig. 3).

3.2.3 Three uses. Correspondence is used in three ways in this formula. (i) The writing side aligns with the reading side: the queries come from the output of the part before it in the same layer of the writing side, and the keys and values from the output of the reading side, so that every mark written can see all the marks read. (ii) The reading side corresponds within itself: queries, keys and values all come from the layer below on the reading side, and every mark can see all the marks on that side. (iii) The writing side corresponds within itself in the same way, but may not look ahead: before sharing, the weighing of the i-th query against every key after i is set to minus infinity.

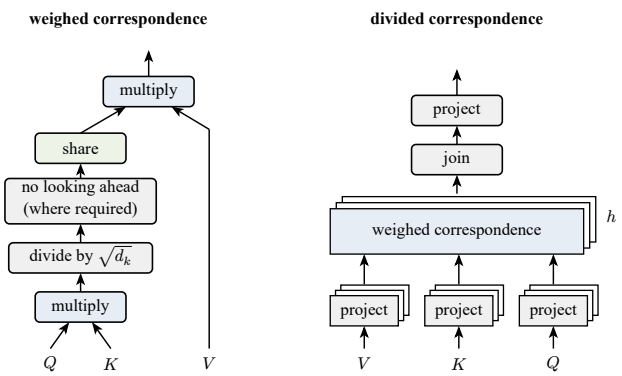


FIG. 2. Left, weighed correspondence. Right, divided correspondence, made of h weighed correspondences in parallel.

3.3 Position-wise transform

Besides correspondence, every layer has a position-wise transform: the row of each mark is projected twice, separately, and the negative numbers are clipped between:

$$\mathrm{transform}(x) = \mathrm{clip}(xW_1 + b_1)W_2 + b_2 \tag{4}$$

The inner width is 2048. The same weights serve every position; each layer has its own.

3.4 Rows of marks, and sharing

Each mark is first turned into a row of width d by a learned table, and multiplied by $\sqrt{d}$ before the position marks are added; the output of the writing side is turned by one projection and a share into the chances of the next mark. The reading side, the writing side and the output use one and the same table, since both sides use the same code.

3.5 Position marks

Correspondence by itself takes no account of order: if a sequence of marks is shuffled, the outputs are merely shuffled with it, and what each mark receives is unchanged. A position mark must therefore be added to the row of every mark. We follow the manner of the tide-tables, which write the height of the tide as a sum of constituents of different periods [12]; we too give each position a set of sines and cosines of different periods,

$$\mathrm{pos}(p, 2i) = \sin\!\big(p / 10000^{2i/d}\big), \quad \mathrm{pos}(p, 2i+1) = \cos\!\big(p / 10000^{2i/d}\big) \tag{5}$$

where i runs from 0 to $d/2 - 1$, the periods running in geometric progression from 2π to about $10000 \cdot 2\pi$. Between the position marks of two positions a fixed distance apart there is then a fixed transformation, so that correspondence can easily learn relations of the kind "so many marks away". Learned position marks [6] did about as well (Table 4, row I); we keep the sines, because they can be written for passages longer than any seen in training: the training passages were mostly under a thousand marks, and on test passages of one to two thousand marks the sines did 0.8 points better than learned marks.

3.6 Beyond the range: cross-volume retrieval

The working range is fixed at 2048 marks, of which the text itself, with the task mark, has at most 1536, the remaining 512 being kept for material brought in. When the transcription or trace of a damaged book lies outside the text, it is brought in by the catalogue retrieval of the Great Library [13], and the reading is done twice. The first reading reads the text alone. Then the outputs of the top layer of the reading side, averaged over the positions and projected by a further array W^R of $d \times d_k$, give a single retrieval query, which is weighed against the head keys of the catalogue; the four passages of highest raw score (the inner product, neither divided by $\sqrt{d_k}$ nor shared) are taken, each cut to 128 marks in all, counting the outer-volume mark, bearing its volume number, that precedes it, and joined after the text. The second reading reads the text and the four passages together from the beginning, the passages corresponding like the text, and their position marks continuing those of the text. The head key of a passage is made by reading its first 64 marks with the reading side, averaging in the same way, and projecting the result by the same W^R; in training the keys are computed afresh every ten thousand steps with the weights of the moment, and stay fixed in between. Retrieval is not adjusted together with correspondence: W^R is trained apart, on the cross-volume correspondences known in the parallel copies, the passage known to

TABLE 1. Three ways compared.

Way	*Cost of a layer*	*Steps in order*	*Most steps between two marks*
Correspondence	$n^2 \cdot d$	1	1
Carried reading	$n \cdot d^2$	n	n
Windows (spacing doubled)	$k \cdot n \cdot d^2$	1	$\log_k n$
Correspondence with the nearest r only	$r \cdot n \cdot d$	1	n/r

hold the transcription of a code-name being counted right and the rest wrong. The catalogue holds only those parts of the repository that this work was permitted to read; sealed and licensed parts are left out under the present grades of holding and rules of leave.

4 Why correspondence

Table 1 compares correspondence, carried reading and windows on three counts: the cost of each layer; the steps within a layer that must be taken in order, which decide how many paths can run in parallel; and the most steps the information between two marks must pass, which decides whether distant relations are easily learned. n is the length of the range, d the width, k the width of the window.

Table 1 compares the core step of each kind of layer, constants neglected; a layer of correspondence has besides its projections and its position-wise transform, which cost about $12n \cdot d^2$ when the inner width is $4d$, and retrieval is reckoned apart. In damaged books n is mostly between some hundreds and two thousand. By the expressions of Table 1, at $n = 2048$ and $d = 512$ correspondence itself costs four times as much as carried reading; with the projections and the transform counted, a layer of correspondence still costs less than three times a layer of gated carried reading [4], some $8n \cdot d^2$, and need not be done in order. Were the range greater still, each mark could be made to attend only to the r nearest it; this we leave for later.

Correspondence has a further merit: it can be seen. The weights of every path can be printed and read, showing at a glance what is being read and where it is matched (Fig. 3). The carried note of carried reading is only a row of numbers.

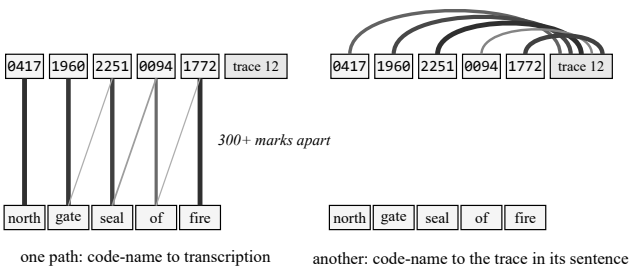


FIG. 3. The weights of two paths in the fifth layer of divided correspondence. Above, a sentence of the volume: five code-names and the trace number at its end; below, the transcription of those five code-names elsewhere in the same volume, more than three hundred marks away. The thicker the line, the greater the weight. The left path joins each code-name to its transcription; the right joins the code-names to the trace number in the same sentence.

5 Training

5.1 Material and tasks. The Repository Office of the Great Library gave us two sets of parallel copies. One is of code-names and transcriptions, 4.1 million pairs: the reading side reads a passage of a damaged book, holding code-names and also scattered, broken pieces of their transcriptions, and the writing side writes the full transcription of every code-name in the passage, in the order of the code-names, each closed by a separating mark, a code-name it cannot answer receiving a mark of doubt. The transcriptions themselves are often broken, and cannot simply be copied, but must be put together from several pieces and the text around them. The "distance" of Table 3 is the number of marks, in the original volume, between a code-name and the nearest piece of its transcription; where it exceeds the range of the text, or the pieces lie only in another volume, they can be seen only through retrieval, and Table 3 counts them together as beyond the range. The other set is of transcriptions and casting traces, 1.2 million pairs: the reading side reads a transcription, and the writing side writes the number of the trace and the version of the formula it answers to; the source and version of every trace were checked by its signature [14], and whether its content is true was examined by the collators apart. The two tasks were trained together, a task mark being set before the reading side. Of the thirty-six thousand characters of the Code Table, some eleven thousand occur in the parallel copies; when those that often occur together had been merged into single units [15], the table of units held thirty-two thousand entries, rare characters being written out one by one in their codes, so that every text can be restored without loss. The material was put in batches of like length, each of about twenty-five thousand marks over the eight paths together. For each task eight thousand pairs were set aside for testing and four thousand for tuning; the division was made by whole volumes, so that variant copies of a volume and repeated transcriptions fall on one side only. In testing, the catalogue was fixed at its version of F.E. 349, and the collated copies used for testing were not entered in it.

5.2 Engines and schedule. We trained in the engine room of the Kanaich Institute of Technology on eight paths; an engine-room day in this paper means those eight paths fully used for one day. The base formula is that of § 3; the large formula has width $d = 1024$, sixteen paths each of width 64, an inner width of 4096 in the position-wise transform, and dropping at three-tenths. The engine-room days of Table 2 include retrieval and the recomputing of head keys, but not the first making of the catalogue. Each step of the base formula took about 0.4 seconds, 80,000 steps in all, about nine hours; each step of the large formula about 1.2 seconds, 250,000 steps, about three and a half days.

5.3 Adjustment. The loss is the negative logarithm of the chance given to the right next mark, and the weights begin as small random numbers. Weights were adjusted by a method of gradients estimated step by step [16], with rates of decay 0.9 and 0.98, the step-size rising over the first four thousand steps and thereafter falling as the inverse square root of the step number:

$$\mathrm{rate} = d^{-0.5} \cdot \min(\mathrm{step}^{-0.5}, \mathrm{step} \cdot 4000^{-1.5}) \tag{6}$$

5.4 Against over-training. In training, every number in the output of every part was dropped at random with chance one-tenth before being added to the bypass, which we call dropping [17]; nothing is dropped in reading and writing. Nor is the target taken as wholly certain:

Table 2. Letter agreement on two reading tasks, and engine-room days of training.

Formula	*Code-name to transcription*	*Transcription to trace*	*Engine-room days*
Windows [6]	61.9	53.4	6.8
Carried reading with alignment [5]	63.2	54.1	9.6
Carried reading with alignment, with retrieval	64.1	55.0	11.2
Carried reading with alignment, seven together	64.5	56.0	61
Correspondence, base	66.8	58.7	0.4
Correspondence, large	68.9	61.3	3.5

the right unit is given nine-tenths and the remaining tenth is spread over the other units, which we call sparing [18]. Sparing makes the chances written less certain, but raises the letter agreement.

6 Results

6.1 Reading damaged books. Table 2 gives the letter agreement on the eight thousand pairs set aside for each of the two reading tasks: the transcription or trace written is turned into the collated copy by the fewest insertions, deletions and changes, and the number of these, divided by the letters of the collated copy, is taken from one and counted in the hundred; letters written in excess must be deleted, and cost as much as any other. Every formula was trained afresh on our material, at a size near that of the base formula. Carried reading with alignment [5] is the best single formula now in use, and is listed also with the same retrieval; "seven together" is seven formulae of carried reading with alignment voting together, not all of the same size. Engine-room days are reckoned as in § 5.2.

The large formula of correspondence is 4.4 and 5.3 points above the seven together on the two tasks, and needs only about a seventeenth of their engine-room days. Carried reading with alignment gains only 0.9 points on each task from the same retrieval: the advantage of correspondence does not lie in retrieval alone.

The difference is greatest for correspondences far apart. Table 3 sorts the correspondences of the first task by the distance between code-name and transcription, and gives the part rightly matched: the transcription written for a code-name counts as matched if more than half of its letters agree with the collated transcription of that code-name, and it agrees with that transcription better than with that of any other code-name in the volume; no answer, or a match to another code-name, counts as wrong. Letter agreement measures how well the letters are written; this measures whether the right place was found.

Of correspondences from two hundred to 1536 marks apart, carried reading with alignment matches fewer than four in ten, correspondence more than eight. Beyond the range, without retrieval, both match only about one in ten; with it, correspondence matches more than six. Those matched without retrieval are mostly common figures whose transcriptions were met in training.

Table 3. Correct correspondence by distance (code-name to transcription, the eight thousand pairs set aside).

Distance of transcription from code-name	*Code-names*	*Carried reading with alignment*	*Correspondence, large, no retrieval*	*Correspondence, large*
Within 50 marks	41,200	91%	93%	93%
51 to 200	18,600	74%	89%	89%
201 to 1536	9,300	38%	84%	85%
Beyond the range (incl. other volumes)	3,900	9%	11%	63%

6.2 Variations. To see the use of each part, we altered the base formula one item at a time and compared the results on the four thousand pairs kept for tuning (Table 4); every variation was trained twice and the two runs averaged, the runs never differing by more than 0.3 points. In every row d_v equals d_k, and the inner width of the position-wise transform is four times d.

A single path (A) is nearly two points below eight; too many paths (C) are no better. Halving d_k (D) also loses. More layers and greater width both help (F, G); without dropping (H) the letter agreement on the training material is higher than for the base formula but lower on the tuning pairs, which shows over-training. Learned position marks (I) differ little from sines. Without retrieval (J) nearly two points are lost, almost all of them in correspondences beyond the range: on the tuning pairs, the base formula's correct correspondence beyond the range falls from 58 to 10 in the hundred.

6.3 Parsing old formulae. The correspondence formula serves other tasks as well. A rune-cutter reading a formula for an old engine type must first parse it: which units make up a loop, which stretch is the two arms of a branch. We trained on forty thousand old formulae parsed by hand by rune-cutters of the Guild of Rune Mages, and tested on two thousand more (Table 5). Each part found, a loop or one arm of a branch, counts as right if it begins

Table 4. Variations of the correspondence formula (code-name to transcription, the four thousand tuning pairs).

Variation	*Paths h*	d_k	*Layers N*	*Width d*	*Dropping*	*Letter agreement*
Base	8	64	6	512	0.1	65.9
A	1	512	6	512	0.1	64.0
B	4	128	6	512	0.1	65.4
C	32	16	6	512	0.1	65.2
D	8	32	6	512	0.1	65.1
E	8	64	2	512	0.1	62.7
F	8	64	8	512	0.1	66.3
G	8	64	6	1024	0.1	67.2
H	8	64	6	512	0	64.6
I. Learned position marks	8	64	6	512	0.1	65.8
J. No retrieval	8	64	6	512	0.1	64.1

Table 5. Parsing old formulae.

Method	*Agreement*
The rune-cutters' table of rules [19]	88.3
Carried reading with alignment [5]	89.6
Correspondence, four layers	91.2

and ends where the rune-cutters' part does; agreement is twice the number of right parts divided by the number found and the number of the rune-cutters' parts together, counted in the hundred. A parse that does not form a lawful structure counts wholly wrong.

The correspondence formula was not altered for this task, except that the stack was reduced to four layers.

6.4 How one damaged book was read. Tables 2 to 5 are statistics; here we follow one volume, to see how the correspondence formula actually reads. The third volume of the Record of the North Gate Seal in the Great Library is a seal journal burned in the Great Tide, each entry of which notes the number of its trace. It holds more than nineteen hundred code-names; their transcriptions are scattered between this volume and the seventh, and the records of the casting traces are in the trace-rolls of the seal-store; the Repository Office had set it aside, unread, for more than a century. In reading it, the large formula placed the greatest weight of one path in its fifth layer, for the forty-first code-name of the volume, on a transcription three hundred and seventy marks further on (Fig. 3, left); another path joined the same code-name to the number of the trace in the same sentence (Fig. 3, right), and the passage of the trace-roll brought in by retrieval records the caster of that seal and the version of his formula. Of the 1,927 code-names in the volume, the large formula wrote transcriptions for 1,618 and gave no answer for 309; two collators of the Repository Office checked each one separately, a third deciding the 23 on which they differed, and 1,540 agreed, the errors lying mostly where the transcription itself is damaged; by the measure below, the volume was read for the greater part. We have since tried the large formula on three hundred of the volumes the Repository Office had set aside: a volume counts as read through when transcriptions accepted by the collators were written for nine in ten of its code-names, and forty-one were so read; one hundred and seventeen more were read for the greater part, that is, for more than half but less than nine in ten.

7 Conclusion

The formula of this paper uses correspondence alone; neither its reading side nor its writing side uses carried reading or windows. In reading damaged books it does better than both, best of all on correspondences far apart, and it trains much faster.

We hope to use it more widely: to read array diagrams and star charts, to read recordings of the speaking-network, and to read over longer ranges. We hope also to use it to read the whole code repository: all the documents, formulae and collations entered in a hundred and thirty years and more, which no scholar could read in a lifetime.

The number of layers, the range, the passages brought in and the longest writing are all fixed, so that the operations of every step of the correspondence formula are bounded beforehand by an amount reckonable from these numbers; the formula therefore always stops in reading and writing a passage, and by Cummings [20] it belongs to a class whose

stopping can be completely judged. This says only that it stops, not that it reads rightly; whether it reads rightly is for the collators to check.

The pattern of correspondence came from the doctrine of seven layers, as has been said. That the formula works does not prove the doctrine. But without what that seer saw deep within the White Tower, we should probably never have thought of sending information all at once, instead of passing it on step by step.

Acknowledgments. The Repository Office of the Great Library gave us the parallel copies and the records of collation; the codes entered letter by letter by the copyists of Copyists' Street are the foundation of everything here. The engine room of the Kanaich Institute of Technology let us use its spare capacity for a year. The repeated objections of several colleagues caused § 3.2 to be rewritten three times.

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18. Skidmore, "Sparing the target", Runes and Formulae 105 (F.E. 345).
19. The Guild of Rune Mages, Rules for Parsing Formulae of the Old Engine Types (F.E. 276).
20. Cummings, "On decidable matters, with an answer to Severin's Fifth Question", Annals of the Laws 223 (F.E. 319).
21. Ashby, Universal Rune Units (Lionwing, c. F.E. 200).

Appendix. Correspondence in universal units

Following the custom of this journal, weighed and divided correspondence are given here in universal units [21]. Any engine type that realises all the universal units, with enough store and speed, may cut them by its transcription table; levelling, sharing and the multiplication of arrays use the standard sub-chains of the supplement to this number, numbers being held to thirty-two places in fixed point. The values of the weights are given in code in the supplement, together with a fixed input and the output it should give, for checking after cutting.

```
-- shut or open: if shut, a mark may not attend to the marks after it; X
and Y may differ in length, i running over X and j over Y
⟨Weighed Q, K, V, shut⟩
  for each i:
    for each j:
      score[i, j] ← Q[i]·K[j] ÷ √d_k
      if shut and j > i: score[i, j] ← the least number fixed point can
hold (masked)
    weight[i] ← share(score[i])
    out[i] ← sum over j of (weight[i, j] · V[j])
  yield out
⟨Divided X, Y, h, shut⟩
  for each path l = 1, …, h (all at once):
    path[l] ← ⟨Weighed X·Wq[l], Y·Wk[l], Y·Wv[l], shut⟩
  yield join(path[1], …, path[h]) · Wo
⟨One layer, reading side X⟩
  X ← level(X + ⟨Divided X, X, 8, open⟩)
  X ← level(X + transform(X))
  yield X
⟨One layer, writing side Y, Z⟩
  Y ← level(Y + ⟨Divided Y, Y, 8, shut⟩)
  Y ← level(Y + ⟨Divided Y, Z, 8, open⟩)
  Y ← level(Y + transform(Y))
  yield Y
```